SYSDESIGN REFERENCE

Design a Search Autocomplete

Requirements-to-architecture reference walkthrough.

Return low-latency prefix suggestions ranked by relevance and freshness.

VERSION 1.0 REVIEW

REVIEWED THROUGH 2026-07-18

STATUS HUMAN REVIEW REQUIRED

SYSDESIGN / SEARCH-AUTOCOMPLETE

SYSDESIGN

Design a Search Autocomplete

CORE IDEA

Return low-latency prefix suggestions ranked by relevance and freshness.

Reference architecture decision chain

01	02	03	04
Trie Or Search Index	Aggregation Pipeline	Personalization	Typo Handling

Decisions to defend

- Trie or search index
- aggregation pipeline
- personalization
- typo handling
- and cache tiers.

Failure modes to test

- Offensive suggestions
- trend spikes
- stale rankings
- and high-cardinality prefixes.

Interview flow

- [ ] Clarify users, critical paths, and non-goals
- [ ] Estimate only scale that changes a decision
- [ ] Define API, data model, and source of truth
- [ ] Draw synchronous and asynchronous paths
- [ ] Pressure-test timeout, retry, overload, and recovery
- [ ] Close with tradeoffs and one alternative

Deep-dive drill

Design a Search Autocomplete

Probe	What a strong answer makes explicit
How does the design behave during a regional dependency failure?	Requirement, decision, cost, failure behavior, and observable metric.
Which metric reveals the first bottleneck?	Requirement, decision, cost, failure behavior, and observable metric.
What consistency does the user actually observe?	Requirement, decision, cost, failure behavior, and observable metric.
How would the design change at ten times the traffic?	Requirement, decision, cost, failure behavior, and observable metric.

Data questions

+ Partition key
+ Versioning
+ Uniqueness
+ Retention
+ Reconciliation

Operations questions

+ SLO
+ Backlog
+ Capacity
+ Alert
+ Failover test

DESIGN DISCIPLINE

For every box: which requirement justifies it, what cost does it introduce, and what happens when it fails?

PUBLISHING STATUS

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